

Endometriosis Across the Menstrual Cycle

Investigating Phase-Related Symptom Variation

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1 Introduction

The menstrual cycle comprises four phases—menstrual, follicular, ovulatory, and luteal—characterized by distinct hormonal profiles and associated physiological changes. Endometriosis is a chronic inflammatory gynecological disease marked by the presence of endometrial-like tissue outside of the uterus, accompanied by painful symptoms and co-occurrence with other health comorbidities [3]. It affects approximately 10% of women of reproductive age, but patients face prolonged delays between the onset of symptoms and diagnosis due to normalization of menstrual pain and reliance on histological verification [1].

This project aimed to investigate how general physical state together with gastrointestinal (GI) issues and pain level—two common manifestations of endometriosis [1]—change across the menstrual cycle. A better understanding of the disease and its impact on everyday life is particularly important in light of gender-related biases in healthcare and gaps in research. The following questions were addressed:

1. Does the **general physical state** differ significantly across the menstrual cycle phases?
2. Do **gastrointestinal (GI) and pain burden intensities** differ significantly across the menstrual cycle phases?
3. Is the general physical state **associated** with GI and pain burden intensities?

2 Dataset

The cycle tracking data were obtained from the published *KI EndoLIST: Endometriosis Longitudinal Individualized Symptoms Tracking Dataset* [6]. The study was conducted on 34 Isreali women with a formal endometriosis diagnosis. Data from each participant was stored in separate CSV files, with rows representing tracking days and columns representing symptoms. Participants could customize both the symptoms tracked and their severity scales, creating individualized profiles. This posed certain challenges for comparative analysis, which were addressed in the preprocessing steps.

- **Participant selection:** Only participants who tracked at least two complete menstrual cycles were included (27/34).
- **Score normalization:** Different severity scales (0–4, 1–4, 1–10) were standardized to a 0–1 range.
- **Cycle phase estimation:** Menstrual, follicular, ovulatory, and luteal phases were assigned using recorded menstruation dates and literature-based assumptions [2, 4]. Exact approach is described in Appendix A.
- **Burden classification:** Individual symptoms were grouped into categories. The complete list of symptoms is provided in Appendix B.

3 Analysis

3.1 General physical state

Normalized daily physical-state scores were averaged within each cycle phase, then across cycles for each participant. Participant-level phase means were used to calculate group averages and paired differences between phases (see Fig. 1). The general physical state was lower during menstruation than during the other phases, with differences ranging between 1σ and 2σ from zero. The differences among non-menstrual phases were less than 1σ away from zero and were hence considered statistically insignificant.

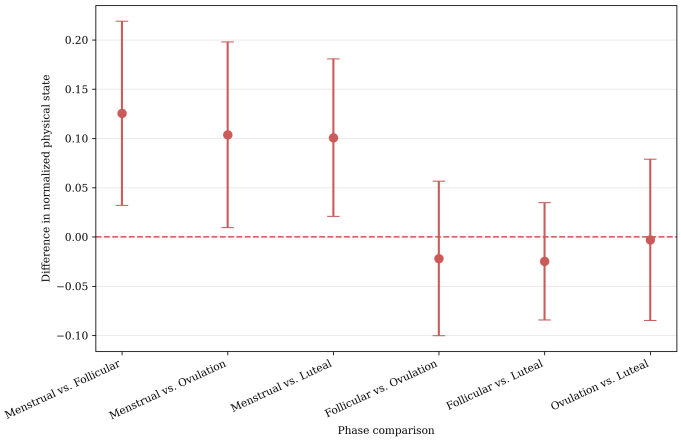
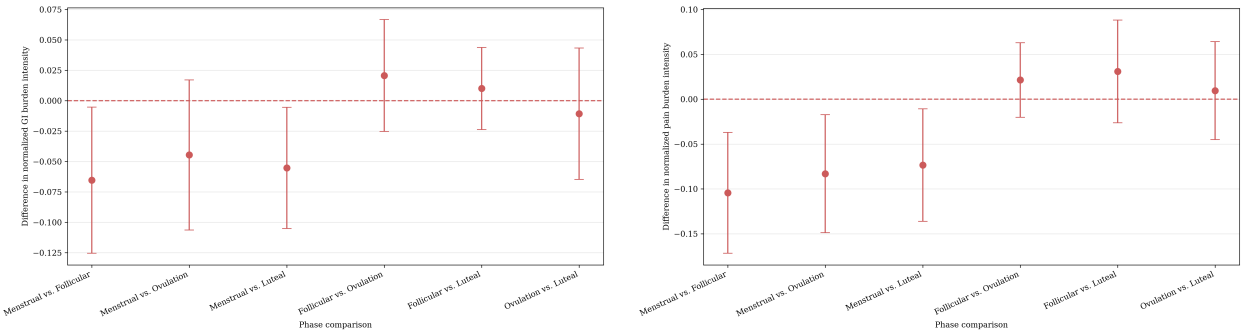


Figure 1: **Differences in the reported general physical state between cycle phases.** Data points represent the mean score differences (second phase - first phase) across the 27 participants. Error bars correspond to the standard deviations of the 27 paired differences.

3.2 GI and pain burden

Daily burden intensity was calculated as the mean normalized intensity of reported symptoms within each category on a given day. The remaining analysis followed the same workflow as for general physical state: averaging across days within each cycle phase, across cycles within participants, and calculating paired phase differences. The GI burden was higher during menstruation than during the follicular and luteal phase (see Fig. 2a). The pain burden was higher during menstruation than during all the other phases (see Fig. 2b). The differences ranged between 1σ and 2σ from zero. The differences among non-menstrual phases were considered statistically insignificant.



(a) GI burden

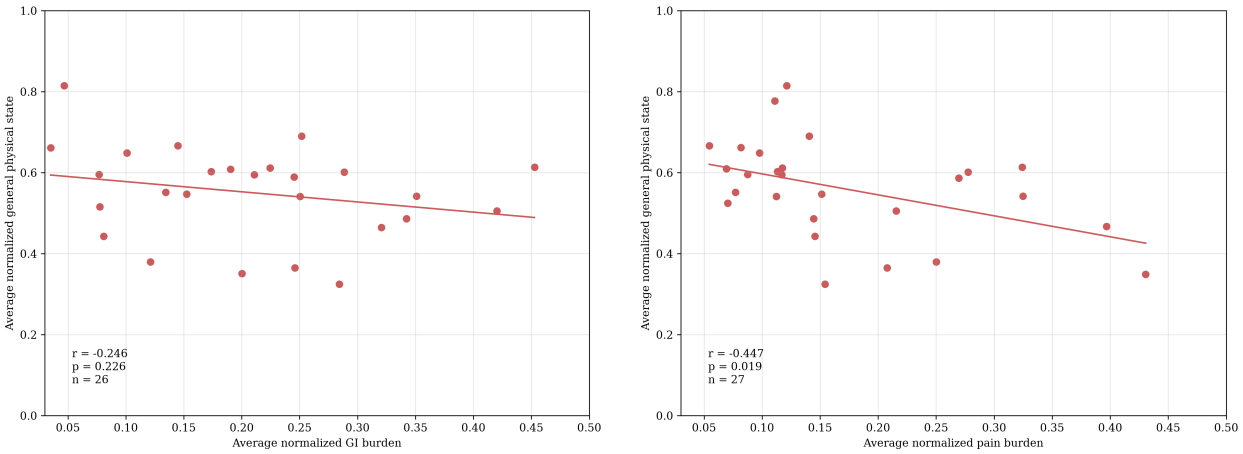
(b) Pain burden

Figure 2: **Differences in the GI and pain burden intensities between cycle phases.** Data points represent the mean score differences across the 27 participants. Error bars correspond to the standard deviations of the 27 paired differences for each burden.

3.3 Correlation

Daily mean GI and pain burden scores were matched with general physical state scores by participant and tracking day. Only days with both measurements available were retained. Scores were then averaged across matched days for each participant, and Pearson's correlation was calculated across participant-level means. Pain burden showed a moderate negative correlation with general physical state ($r = -0.447$, $p = 0.019$, $n = 27$), indicating that higher pain burden was associated with poorer physical state (see Fig. 3b).

This association was statistically significant ($p < 0.05$). In contrast, GI burden showed a weaker negative correlation (r = -0.246, p = 0.226, n = 26), which was not statistically significant (see Fig. 3a).



(a) GI burden

(b) Pain burden

Figure 3: **Associations between symptom burden and general physical state.** Each point represents one participant, with values averaged across days with available paired measurements. Lines indicate linear regression fits.

4 Conclusion

General physical state was lowest, while GI and pain burdens were highest during menstruation, although none of the phase differences exceeded the 3σ significance threshold. A significant moderate negative correlation between pain burden and general physical state suggests that greater pain burden is associated with poorer physical state in women with endometriosis. The underlying reason for the observed pattern is that endometriosis symptoms tend to intensify during menstruation [5]. Limitations of the presented analysis include **inconsistent symptom tracking** across participants, **small sample size** of 27 participants, **cycle phase estimates** based on literature rather than hormonal levels, and **grouping diverse symptoms** into burden categories.

References

[1] Sanjay K. Agarwal et al. "Clinical diagnosis of endometriosis: a call to action". In: *American Journal of Obstetrics and Gynecology* 220.4 (2019), 354.e1–354.e12. DOI: 10.1016/j.ajog.2018.12.039.

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[6] Tamar Zelovich et al. "KI EndoLIST: Endometriosis Longitudinal Individualized Symptoms Tracking Dataset". In: *PhysioNet* (Apr. 2026). Version 1.0.0. DOI: 10.13026/k99q-fm63. URL: https://doi.org/10.13026/k99q-fm63.

A Cycle phase estimates

Menstrual cycle phases were estimated using recorded menstruation dates and literature-based assumptions. The luteal phase was used as a reference because its duration is reported to be more consistent than that of the follicular phase [4]. Ovulation window was estimated to last two days [2].

Each tracking day was assigned to one of four phases.

- **Menstrual:** All days recorded as menstruation (Period = 1).
- **Follicular:** First day after menstruation when Period=0 (no) until the day before of the ovulation window.
- **Ovulation:** Two days before the start of the luteal phase.
- **Luteal:** Starts 14 days before the first day of menstruation.

Table 1: Example of phase assignment for a 30-day menstrual cycle.

Cycle days	**Assigned phase**	**Notes**
1–5	Menstruation	Observed Period = 1
6–14	Follicular	
15–16	Ovulation window	
17–30	Luteal	14 days

B Burden classification

- Example symptoms classified as **GI burden:** bloating, diarrhea, vomiting, stomach discomfort, painful bowel movement.
- Example symptoms classified as **pain burden:** migraines, breast tenderness, pelvic pain, ovaries pain, uterine cramps.